

Huang-Cheng Chou

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Postdoctoral Scholar ([NSTC Fellow](#)) at [USC SAIL](#) with [Prof. Shrikanth S. Narayanan](#). Speech emotion recognition under subjectivity, speech / audio LLMs, fairness and calibration, and [speech enhancement for real-time MRI](#).

715 citations h-index 14 i10-index 23 [Google Scholar, 18 Aug 2026](#)

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Summary

I am a Postdoctoral Scholar– [NSTC Fellow](#) at the [Signal Analysis and Interpretation Laboratory \(SAIL\)](#) at the [University of Southern California](#), working with [Prof. Shrikanth S. Narayanan](#), including the [Speech Production and Articulation Knowledge \(SPAN\)](#) effort. I received my Ph.D. in Electrical Engineering from [National Tsing Hua University \(NTHU\)](#), advised by [Prof. Chi-Chun Lee \(BIIC Lab\)](#). My dissertation, *Revisiting Modeling and Evaluation Approaches in Speech Emotion Recognition: Considering Subjectivity of Annotators and Ambiguity of Emotions*, received the [ACLCLP](#) Doctoral Dissertation Award, Honorable Mention.

My work treats emotion labels as subjective, multi-label, and often disagreeing human judgments rather than a single ground truth. That program runs from annotator modeling and calibration (ICASSP, INTERSPEECH, IEEE TAFEC) through open evaluation ([EMO-SUPERB](#)) to fairness audits and a [survey of bias in speech AI](#). I also work on assistant-style speech systems (unified ASR + SER at [Amazon Alexa](#); on-device SER / KWS at [RealTek](#)), speech-LLM evaluation and resources, and first-author assessment of [speech enhancement for real-time MRI audio](#).

Authorship and contribution notes. Name in **bold** is Huang-Cheng Chou. [EMO-SUPERB / Open-Emotion](#) is equal-first with Haibin Wu. On [DeSTA2.5-Audio](#) I contributed the emotion-data stack (label schema and dimensional annotations), not the full model. On [Do You Hear What I Mean?](#) I led the core evaluation pipeline, not the TTS model. Papers marked Mentored are student-led work I advised. The APSIPA ASC Best Regular Paper Award (2019) is for [conversational deception detection](#), not later SER papers.

Research interests

- Speech emotion recognition under annotator subjectivity and emotion ambiguity: multi-label learning, soft labels, calibration, and evaluation rules that keep minority ratings.
- Speech / audio large language models: evaluation contracts, emotion-aware data, and downstream SER with speech LLMs.
- Fairness, bias, and reliability of speech systems (SER, TTS evaluation, speaker, and cross-task surveys).

- Conversational and dyadic social signal processing: deception, belongingness, and clinical affect such as depression-related speech.
 - Multitask and edge speech systems: unified ASR + SER, keyword spotting, and compute-constrained on-device models.
 - Low-resource and multilingual speech (including [Taigi / Taiwanese Hokkien intent](#)).
 - Speech enhancement and downstream evaluation for [real-time MRI \(rtMRI\)](#) of vocal-tract shaping.
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Education

Ph.D., Electrical Engineering

[National Tsing Hua University \(NTHU\)](#) · Advisor: [Prof. Chi-Chun Lee \(BIIC Lab\)](#)

Feb 2016 – Jul 2024

Hsinchu, Taiwan

- GPA 3.82 / 4.0.
 - Dissertation: *Revisiting Modeling and Evaluation Approaches in Speech Emotion Recognition: Considering Subjectivity of Annotators and Ambiguity of Emotions*.
 - [ACLCLP](#) Doctoral Dissertation Award, Honorable Mention (2024).
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Visiting Scholar (during Ph.D.)

[The University of Texas at Dallas](#) · [MSP Lab](#) · [Prof. Carlos Busso](#) · [NSTC / MOST](#) Graduate Students Study Abroad Grant

Aug 2021 – Sep 2022

Dallas, TX, USA

- Perceptual annotation modeling, calibration, and robustness for speech emotion recognition; collaboration on [MSP-Podcast](#) and related SER evaluation work.
 - This is a visiting appointment during the [NTHU](#) Ph.D., not a separate [UTD](#) degree.
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B.S., Electrical Engineering

[National Tsing Hua University \(NTHU\)](#)

Sep 2011 – Jun 2016

Hsinchu, Taiwan

Appointments

Postdoctoral Scholar (NSTC Fellow)

[University of Southern California](#) · [SAIL](#) / [SPAN](#) · Advisor: [Prof. Shrikanth S. Narayanan](#)

Oct 2025 – Present

Los Angeles, CA, USA

- [NSTC](#) Postdoctoral Research Abroad Program fellow at [SAIL](#).
- First-author work on speech enhancement for real-time MRI: evaluates Denoiser, PASE, and RE-USE across five rtMRI corpora on predicted quality, source preservation, ASR, speaker, and paralinguistic endpoints; preprint [arXiv:2608.16125](#) (submitted to JASA), with an open demo at [rmridemo.huangchengchou.com](#).
- Speech emotion with speech LLMs; mentoring on [VoxEmo](#) (reproducible speech-LLM SER toolkit; IEEE TAFFC, under review).
- Non-verbal vocalizations and speaker identity; cross-lingual speaker verification ([AMECxCxSV](#)); dyadic clinical / depression-related speech (mentored papers).
- Collaborations continuing from the [NTU SPML](#) period, including [DeSTA2.5-Audio](#), [TaigiSpeech](#), and the [fairness survey](#).

Independent Researcher / Collaborator

[National Taiwan University](#) · [Speech Processing and Machine Learning \(SPML\) Lab](#) · [Prof. Hung-yi Lee](#)

Sep 2024 – Oct 2025

Taipei, Taiwan

- No official NTU visiting or employment appointment. Independent research while serving Taiwan substitute military service, in collaboration with [SPML Lab](#) members.
- Fairness and debiasing in multi-label SER ([Emo-bias](#), [EMO-Debias](#), [CO-VADA](#); INTERSPEECH 2025 [subgroup-disparity](#) work with fairness-metric gains above 28% at small accuracy cost).
- Instruction-guided expressive TTS evaluation ([Do You Hear What I Mean?](#), ICASSP 2026; evaluation pipeline).
- Low-resource speech: [TaigiSpeech](#) (INTERSPEECH 2026 long paper).
- Emotion-data / label schema contributions to [DeSTA2.5-Audio](#) (IEEE TASLP 2026).
- Joint publications at INTERSPEECH, ASRU, ICASSP, and IEEE TASLP; roles vary (second / co-author / mentored).

Substitute Military Service

[Taipei City Government](#) · Taiwan mandatory substitute service

Sep 2024 – Sep 2025

Taipei, Taiwan

- Completed Taiwan substitute (alternative) military service. Speech research in this period was independent collaboration with [NTU SPML](#), not an official government research post.

Applied ML / DSP Engineer Intern

[RealTek Semiconductor](#) · Emerging Tech / Advanced DSP

Jun 2024 – Sep 2024

Hsinchu, Taiwan

- Streaming on-device speech emotion recognition and keyword spotting (Python, TensorFlow) under FLOPs / MACs / parameter budgets for consumer devices.
- Compressed SER models with about 30% smaller footprint while targeting the same accuracy operating point; estimated compute for neural KWS and SER on edge runtimes.

Applied Scientist Intern

[Amazon AGI](#) · Alexa Speech / Hybrid Science / ASR

Aug 2023 – Nov 2023

Pittsburgh, PA, USA

- Emotion-aware spoken language understanding for assistant traffic: unified shared-layer ASR + multi-label SER, about 20% lower system complexity than separate ASR and SER stacks.
- Diagnosed Whisper ASR degradation on emotional speech (about 15% higher error versus neutral). Two-stage fine-tuning improved WER by 8.85% absolute (not a relative 29.75% figure).
- Related first-author paper: [Tiny Whisper-SER](#) (APSIPA ASC 2024).

Graduate Research Assistant

[NTHU](#) · [BIIC Lab](#) · Advisor: [Prof. Chi-Chun Lee](#)

Feb 2016 – Jul 2024

Hsinchu, Taiwan

- Speech emotion recognition under annotator subjectivity and emotion ambiguity: multi-label learning, co-rater training, calibration, and all-inclusive aggregation for evaluation.
- Led / co-led Mandarin dyadic corpora [NNIME](#) (interactive multimodal emotion) and [DDDM](#) (deception in dialog games) as community data resources.
- Co-led [EMO-SUPERB / Open-Emotion](#) (IEEE SLT 2024; equal-first with Haibin Wu): standardized splits, codebase, and live leaderboard.
- Mentored junior students on experiments and publications in SER, deception, and small-group conversation.

Research Assistant – automatic meeting summarization

[Industrial Technology Research Institute \(ITRI\)](#)

Dec 2018 – Dec 2019

Hsinchu, Taiwan

- Transformer / BERT pipelines for multi-party spoken meeting summarization.

Research Intern

[Industrial Technology Research Institute \(ITRI\)](#)

Jul 2015 – Sep 2015

Hsinchu, Taiwan

- Polarity-term discovery with NLP over web text (BBS, Yahoo, Twitter) and parallel processing on Apache Spark.

Undergraduate Research Assistant

[NTHU](#) · [BIIC Lab](#) · [Prof. Chi-Chun Lee](#)

Sep 2014 – Jan 2016

Hsinchu, Taiwan

- Collection of a Taiwan Mandarin interactive multimodal emotion corpus ([NNIME](#)) and early machine-learning baselines for Mandarin multimodal emotion recognition; continued into the Ph.D.

Selected research highlights

Grouped by theme. Full bibliographic entries are in Publications.

rtMRI speech and source preservation

- [First] Systematic assessment of off-the-shelf enhancement (Denoiser, PASE, RE-USE) on five rtMRI corpora: predicted quality does not reliably imply better ASR or source fidelity. Treat enhanced rtMRI audio as a task-specific derivative, not a universal replacement.
[arXiv:2608.16125](#); submitted to JASA.

Subjective & ambiguous SER

- [First] Annotator-aware SER from [Every Rating Matters](#) (ICASSP 2019) through co-rater training, co-occurrence, typed descriptions, [calibration](#) (INTERSPEECH 2023), and [all-inclusive aggregation](#) (IEEE TAFRC 2024; SLT 2024).
- [Equal first] [EMO-SUPERB / Open-Emotion](#): leakage-aware splits, SSL baselines, codebase, and live leaderboard (IEEE SLT 2024).

Fairness, safety, and evaluation of expressive speech

- [Co] [Toward Fair Speech Technologies](#) — survey of bias and fairness in speech AI (arXiv:2605.01597; submitted to TMLR), with [awesome-speech-fairness](#) and a [Hugging Face collection](#).
- [Second] [Subgroup-disparity mitigation](#) without requiring explicit demographics (INTERSPEECH 2025); [EMO-Debias](#) (ASRU 2025).
- [Co] [Emo-bias](#), [CO-VADA](#); [Second] instruction-perception gap in expressive TTS ([Do You Hear What I Mean?](#), ICASSP 2026; evaluation pipeline).

Speech / audio LLMs and assistant systems

- [Co] [DeSTA2.5-Audio](#) (IEEE TASLP 2026) — emotion-data / labels, not the full model.
- [First] [Tiny Whisper-SER](#); [Amazon Alexa](#) intern: shared-layer ASR + SER and emotional-speech WER diagnosis.
- [Mentored] [VoxEmo](#) toolkit for speech-LLM SER.

Low-resource, speaker, and clinical speech

- [Co] [TaigiSpeech](#); [under-resourced SER via speech-to-speech translation](#) (ICASSP 2025).
- [Mentored] [Speaker identity in non-verbal vocalizations](#); [AMECxSV](#); [dyadic depression detection](#) papers.

Conversational social signals

- [First] Deception in Mandarin dialog games ([APSIPA ASC 2019 Best Regular Paper](#); [APSIPA TSIP 2021](#); [ACL Findings 2021](#)).
- [First / lead] [NNIME](#) dyadic corpus (ACII 2017); [Second] [belongingness and group emotion](#) at ICMI 2021.

Awards & honors

[NSTC](#) Postdoctoral Research Abroad Fellowship — National Science and Technology Council, Taiwan, 2025–2026

Merry Electronics Electroacoustics Thesis Award — Silver (2025) and Bronze (2021)

[ACLCLP](#) Doctoral Dissertation Award — Honorable Mention (2024)

NOVATEK Ph.D. Excellence Scholarship — 2022–2023

The Rotary Foundation Excellence Scholarship — 2021

MOST ([NSTC](#)) Graduate Students Study Abroad Grant — 2020–2022 ([UT Dallas](#) visiting year)

APSIPA ASC Best Regular Paper Award — 2019 ([conversational deception detection](#))

MOST Futuretek Breakthrough Award — 2019

[NTHU](#) Dean Ph.D. Student Excellence Scholarship — 2016–2020

TITC Excellence Scholarship — 2015

TSMC Excellence Scholarship — 2014

Travel support

IEEE SPS ICASSP Travel Grant — 2025

IEEE SLT Travel Grant — 2024

[ACLCLP](#) Outstanding Students Conference Travel Grant — 2019, 2022, 2024, 2025

FAOS Outstanding Students Conference Travel Grant — 2019, 2022, 2023

Google East Asia Student Travel Grants — ICASSP 2022 and INTERSPEECH 2022

[ISCA](#) INTERSPEECH Grant — 2022

AAAC ACII Student Travel Grant — 2017

Academic service & mentoring

- **Journal reviewer:** IEEE/ACM TASLP; IEEE TAFRC; Elsevier Pattern Recognition; Computer Speech and Language; Information Processing & Management; APSIPA TSIP; IEEE Transactions on Multimedia.
- **Conference reviewer:** INTERSPEECH (2022–2025); ICASSP (2022–2023); ACM Multimedia (2024).
- [ISCA Student Advisory Committee](#) (2023–2024), including student mentoring events.
- IEEE SLT 2024 committee service.
- **Mentoring:** [VoxEmo](#); [AdaLTM](#); [dyadic depression detection](#); [speaker identity in non-verbal vocalizations](#); [AMECXS](#). Junior [BIIC](#) students on SER and conversational experiments during the Ph.D.

Languages

- Mandarin Chinese — native.
 - English — professional working proficiency.
 - Taiwanese (Taigi / Hokkien) — full professional proficiency.
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Open resources

[EMO-SUPERB / Open-Emotion](#) — reproducible SER benchmark, splits, and leaderboard. Code: github.com/ag027592/EMO-SUPERB.

[awesome-speech-fairness](#) — living bibliography for the fairness survey. Hub: [Hugging Face collection](#).

[rtMRI speech enhancement demo](#) — companion to [arXiv:2608.16125](#).

[MSP-Podcast](#) — large naturalistic SER corpus (IEEE TAFCC 2026); paper: [DOI](#).

[NNIME](#) — NTHU-NTUA Chinese interactive multimodal emotion corpus (ACII 2017).

[VoxEmo](#) — mentored reproducibility toolkit for speech-LLM SER.

[emotional TTS labeling instructions](#) — annotation interface and examples for expressive TTS perceptual labels.

Technical skills

SPEECH / ML

ASR, SER, speech LLMs, Whisper / SSL fine-tuning, TTS evaluation, neural codecs, KWS, multilingual and low-resource speech; calibration and fairness evaluation; PyTorch, TensorFlow, [Hugging Face](#).

SYSTEMS

Python, SQL, Git, Linux; AWS / S3 speech-data workflows; Praat, openSMILE, librosa, Fairseq.

EDGE

Streaming inference, on-device FLOPs / MACs / parameter budgets, model compression for SER / KWS.

Publications

Complete list grouped by authorship role. **Huang-Cheng Chou** in bold. Also on [Google Scholar](#) (715 citations; h-index 14; i10-index 23 as of 18 Aug 2026).

First-author (including equal-first)

1. FIRST

Navigating Speech Enhancement for Real-Time MRI: A Systematic Assessment of Signal Quality, Source Preservation, and Downstream Tasks

Huang-Cheng Chou, Sean Foley, Haley Hsu, Kevin Huang, Szu-Jui Chen, Rong Chao, Louis Goldstein, Khalil Iskarous, Dani Byrd, Yu Tsao, Sudarsana Reddy Kadiri, John H. L. Hansen, Shrikanth Narayanan
arXiv:2608.16125, 2026 (submitted to JASA)

[arXiv](#) · [Demo](#)

2. FIRST

Stimulus Modality Matters: Impact of Perceptual Evaluations from Different Modalities on Speech Emotion Recognition System Performance

Huang-Cheng Chou, Haibin Wu, Chi-Chun Lee

ICASSP 2025

[DOI](#)

-
3. FIRST
- A Tiny Whisper-SER: Unifying Automatic Speech Recognition and Multi-label Speech Emotion Recognition Tasks**
- Huang-Cheng Chou
- APSIPA ASC 2024
- [DOI](#)
-

4. FIRST
- Empower Typed Descriptions by Large Language Models for Speech Emotion Recognition**
- Huang-Cheng Chou, Haibin Wu, Kai-Wei Chang, Lucas Goncalves, Jiawei Du, Jyh-Shing Roger Jang, Chi-Chun Lee, Hung-yi Lee
- APSIPA ASC 2024
- [DOI](#)
-

5. EQUAL FIRST
- Open-Emotion: A Reproducible EMO-Superb For Speech Emotion Recognition Systems**
- Huang-Cheng Chou^{*}, Haibin Wu^{*}, Kai-Wei Chang, Lucas Goncalves, Jiawei Du, Jyh-Shing Roger Jang, Chi-Chun Lee, Hung-yi Lee
- IEEE SLT 2024
- *Equal contribution with Haibin Wu.
- [DOI](#) · [Project](#) · [Code](#)
-

6. EQUAL FIRST
- EMO-SUPERB: An In-depth Look at Speech Emotion Recognition**
- Haibin Wu^{*}, Huang-Cheng Chou^{*}, Kai-Wei Chang, Lucas Goncalves, Jiawei Du, Jyh-Shing Roger Jang, Chi-Chun Lee, Hung-yi Lee
- arXiv:2402.13018, 2024 — extended preprint of the IEEE SLT 2024 EMO-SUPERB / Open-Emotion paper, not a separate publication.
- [arXiv](#)
-

7. FIRST
- Embracing Ambiguity And Subjectivity Using The All-Inclusive Aggregation Rule For Evaluating Multi-Label Speech Emotion Recognition Systems**
- Huang-Cheng Chou, Lucas Goncalves, Haibin Wu, Hung-yi Lee, Chi-Chun Lee
- IEEE SLT 2024
- [DOI](#)

-
8. FIRST
- Minority Views Matter: Evaluating Speech Emotion Classifiers with Human Subjective Annotations by an All-Inclusive Aggregation Rule**
- Huang-Cheng Chou, Lucas Goncalves, Seong-Gyun Leem, Ali N. Salman, Chi-Chun Lee, Carlos Busso
IEEE Transactions on Affective Computing, 2024
[DOI](#)
-
9. FIRST
- The Importance of Calibration: Rethinking Confidence and Performance of Speech Multi-label Emotion Classifiers**
- Huang-Cheng Chou, Lucas Goncalves, Seong-Gyun Leem, Chi-Chun Lee, Carlos Busso
INTERSPEECH 2023
[DOI](#)
-
10. FIRST
- Do Minority Views in Perceptual Evaluations Affect Confidence of Speech Emotion Classifiers?**
- Huang-Cheng Chou
ACII Workshops and Demos (ACIIW) 2022
[DOI](#)
-
11. FIRST
- Predicting Inter-annotator Agreements to Improve Calibration and Performance of Speech Emotion Classifiers**
- Huang-Cheng Chou
8th Doctoral Consortium, INTERSPEECH 2022
[PDF](#)
-
12. FIRST
- Exploiting Co-occurrence Frequency of Emotions in Perceptual Evaluations To Train A Speech Emotion Classifier**
- Huang-Cheng Chou, Chi-Chun Lee, Carlos Busso
INTERSPEECH 2022
[DOI](#) · [ISCA](#)
-
13. FIRST
- Exploiting Annotators' Typed Description of Emotion Perception to Maximize Utilization of Ratings for Speech Emotion Recognition**
- Huang-Cheng Chou, Wei-Cheng Lin, Chi-Chun Lee, Carlos Busso
ICASSP 2022
[DOI](#)

-
14. FIRST
- “Does it Matter When I Think You Are Lying?” Improving Deception Detection by Integrating Interlocutor’s Judgements in Conversations**
- Huang-Cheng Chou, Woan-Shiuan Chien, Da-Cheng Juan, Chi-Chun Lee
- Findings of ACL-IJCNLP 2021
- [ACL](#) · [DOI](#)
-
15. FIRST
- Automatic Deception Detection using Multiple Speech and Language Communicative Descriptors in Dialogs**
- Huang-Cheng Chou, Yi-Wen Liu, Chi-Chun Lee
- APSIPA Transactions on Signal and Information Processing, 2021
- [DOI](#)
-
16. FIRST
- “Your Behavior Makes Me Think It Is a Lie”: Recognizing Perceived Deception using Multimodal Data in Dialog Games**
- Huang-Cheng Chou, Chi-Chun Lee
- APSIPA ASC 2020
- [IEEE](#)
-
17. FIRST
- Learning to Recognize Per-Rater’s Emotion Perception Using Co-Rater Training Strategy with Soft and Hard Labels**
- Huang-Cheng Chou, Chi-Chun Lee
- INTERSPEECH 2020
- [DOI](#)
-
18. FIRST
- Joint Learning of Conversational Temporal Dynamics and Acoustic Features for Speech Deception Detection in Dialog Games**
- Huang-Cheng Chou, Yi-Wen Liu, Chi-Chun Lee
- APSIPA ASC 2019 — Best Regular Paper Award
- [DOI](#)
-
19. FIRST
- Every Rating Matters: Joint Learning of Subjective Labels and Individual Annotators for Speech Emotion Classification**
- Huang-Cheng Chou, Chi-Chun Lee
- ICASSP 2019
- [DOI](#)

20. FIRST

NNIME: The NTHU-NTUA Chinese Interactive Multimodal Emotion Corpus

Huang-Cheng Chou, Wei-Cheng Lin, Lien-Chiang Chang, Chyi-Chang Li, Hsi-Pin Ma, Chi-Chun Lee

ACII 2017

[DOI](#)

21. FIRST

Amplifying a Sense of Emotion toward Drama—Long Short-Term Memory Recurrent Neural Network for Dynamic Emotion Recognition

Huang-Cheng Chou, Chun-Min Chang, Yu-Shuo Liu, Shiuan-Kai Kao, Chi-Chun Lee

ROCLING 2017

[ACL](#)

Second-author

1. SECOND

Do You Hear What I Mean? Quantifying the Instruction-Perception Gap in Instruction-Guided Expressive Text-to-Speech Systems

Yi-Cheng Lin, Huang-Cheng Chou, Tzu-Chieh Wei, Kuan-Yu Chen, Hung-yi Lee

ICASSP 2026

Contribution: core evaluation pipeline for the instruction-perception gap, not the TTS model itself.

[DOI](#)

2. SECOND

EMO-Debias: Benchmarking Gender Debiasing Techniques in Multi-Label Speech Emotion Recognition

Yi-Cheng Lin, Huang-Cheng Chou, Yu-Hsuan Li Liang, Hung-yi Lee

IEEE ASRU 2025

[DOI](#)

3. SECOND

Mitigating Subgroup Disparities in Multi-Label Speech Emotion Recognition: A Pseudo-Labeling and Unsupervised Learning Approach

Yi-Cheng Lin, Huang-Cheng Chou, Hung-yi Lee

INTERSPEECH 2025

[DOI](#)

4. SECOND

Jointly Learning From Unimodal and Multimodal-Rated Labels in Audio-Visual Emotion Recognition

Lucas Goncalves, Huang-Cheng Chou, Ali N. Salman, Chi-Chun Lee, Carlos Busso

IEEE Open Journal of Signal Processing, 2025

[DOI](#)

-
5. SECOND
- Contextual Attention for Robust Audio-Visual Emotion Recognition**
Lucas Goncalves, **Huang-Cheng Chou**, Ali N. Salman, Chi-Chun Lee, Carlos Busso
IEEE Open Journal of Signal Processing, 2025
[DOI](#)
-
6. SECOND
- Belongingness and Satisfaction Recognition from Physiological Synchrony with A Group-Modulated Attentive BLSTM under Small-group Conversation**
Woan-Shiuan Chien, **Huang-Cheng Chou**, Chi-Chun Lee
ICMI Companion 2021
[DOI](#)
-
7. SECOND
- Self-assessed Emotion Classification from Acoustic and Physiological Features within Small-group Conversation**
Woan-Shiuan Chien, **Huang-Cheng Chou**, Chi-Chun Lee
ICMI Companion 2021
[DOI](#)
-
8. SECOND
- Acoustic Indicators of Deception in Mandarin Daily Conversations Recorded from an Interactive Game**
Chih-Hsiang Huang, **Huang-Cheng Chou**, Yi-Tong Wu, Chi-Chun Lee, Yi-Wen Liu
INTERSPEECH 2019
[DOI](#)
-
9. SECOND
- Development of a rapid and economic in vivo electrocardiogram platform for cardiovascular drug assay and electrophysiology research in adult zebrafish**
Min-Hsuan Lin, **Huang-Cheng Chou**, Yu-Fu Chen, Wangta Liu, Chi-Chun Lee, Lawrence Yu-Min Liu, Yung-Jen Chuang
Scientific Reports, 2018
[DOI](#)

Co-author

1. CO
- Toward Fair Speech Technologies: A Comprehensive Survey of Bias and Fairness in Speech AI**
Yi-Cheng Lin, Yun-Shao Tsai, Kuan-Yu Chen, Hsiao-Ying Huang, **Huang-Cheng Chou**, Shrikanth Narayanan, Yu Tsao, Jian-Jiun Ding, Hung-yi Lee
arXiv:2605.01597, 2026 (submitted to TMLR)
[arXiv](#) · [Code](#) · [Hub](#)

2. co

DeSTA2.5-Audio: Toward General-Purpose Large Audio Language Model With Self-Generated Cross-Modal Alignment

Ke-Han Lu, Zhehuai Chen, Szu-Wei Fu, Chao-Han Huck Yang, Sung-Feng Huang, Chih-Kai Yang, Chee-En Yu, Chun-Wei Chen, Wei-Chih Chen, Chien-yu Huang, Yi-Cheng Lin, Yu-Xiang Lin, Chi-An Fu, Chun-Yi Kuan, Wenze Ren, Xuanjun Chen, Wei-Ping Huang, En-Pei Hu, Tzu-Quan Lin, Yuan-Kuei Wu, Kuan-Po Huang, Hsiao-Ying Huang, **Huang-Cheng Chou**, Kai-Wei Chang, Cheng-Han Chiang, Boris Ginsburg, Yu-Chiang Frank Wang, Hung-yi Lee

IEEE Transactions on Audio, Speech and Language Processing, 2026

Contribution: emotion-data stack (label schema and dimensional annotations), not the full model architecture.

[DOI](#) · [Model](#)

3. co

TaigiSpeech: A Low-Resource Real-World Speech Intent Dataset and Preliminary Results with Scalable Data Mining In-the-Wild

Kai-Wei Chang, Yi-Cheng Lin, **Huang-Cheng Chou**, Wenze Ren, Yu-Han Huang, Yun-Shao Tsai, Chien-Cheng Chen, Yu Tsao, Yuan-Fu Liao, Shrikanth Narayanan, James Glass, Hung-yi Lee

INTERSPEECH 2026 (long paper); [arXiv:2603.21478](#)

[arXiv](#)

4. co

The False Resonance: A Critical Examination of Emotion Embedding Similarity for Speech Generation Evaluation

Yun-Shao Tsai, Yi-Cheng Lin, **Huang-Cheng Chou**, Tzu-Wen Hsu, Yun-Man Hsu, Chun Wei Chen, Shrikanth Narayanan, Hung-yi Lee

[arXiv:2604.26347](#), 2026

[arXiv](#)

5. co

The Binding Effect: Analyzing How Multi-Dimensional Cues Form Gender Bias in Instruction TTS

Kuan-Yu Chen, Yi-Cheng Lin, Po-Chung Hsieh, **Huang-Cheng Chou**, Chih-Fan Hsu, Jeng-Lin Li, Hung-yi Lee, Jian-Jiun Ding

[arXiv:2603.20743](#), 2026

[arXiv](#)

6. co

AV-EMO-Reasoning: Benchmarking Emotional Reasoning Capabilities in Omni-modal LLMs with Audio-visual Cues

Dingkun Zhou, Krish Patel, Ajay Kankipati, Akshaj Gupta, Zeyi Austin Li, Mohul Shukla, Vibhor Narang, Sara Kofman, Zongli Ye, Grace Wang, Xiaoyu Shi, Tingle Li, Guan-Ting Lin, Kan Jen Cheng, **Huang-Cheng Chou**, Jiachen Lian, Gopala Anumanchipalli

[arXiv:2510.07355](#), 2025

[arXiv](#)

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7. CO
- The MSP-Podcast Corpus**
- Carlos Busso, Reza Lotfian, Kusha Sridhar, Ali N. Salman, Wei-Cheng Lin, Lucas Goncalves, Srinivas Parthasarathy, Abinay Reddy Naini, Seong-Gyun Leem, Luz Martinez-Lucas, **Huang-Cheng Chou**, Pravin Mote
- IEEE Transactions on Affective Computing, 2026 (early access)
- [DOI](#) · [arXiv](#) · [Corpus](#)
-
8. CO
- CO-VADA: A Confidence-Oriented Voice Augmentation Debiasing Approach for Fair Speech Emotion Recognition**
- Yun-Shao Tsai, Yi-Cheng Lin, **Huang-Cheng Chou**, Hung-yi Lee
- IEEE ASRU 2025
- [DOI](#)
-
9. CO
- EMO-Reasoning: Benchmarking Emotional Reasoning Capabilities in Spoken Dialogue Systems**
- Jingwen Liu, Kan Jen Cheng, Jiachen Lian, Akshay Anand, Rishi Jain, Faith Qiao, Robin Netzorg, **Huang-Cheng Chou**, Tingle Li, Guan-Ting Lin, Gopala Anumanchipalli
- IEEE ASRU 2025
- [DOI](#)
-
10. CO
- Meta-PerSER: Few-Shot Listener Personalized Speech Emotion Recognition via Meta-learning**
- Shi-Xin Fang, Liang-Yeh Shen, Yi-Cheng Lin, **Huang-Cheng Chou**, Hung-yi Lee
- INTERSPEECH 2025
- [DOI](#) · [Code](#)
-
11. CO
- Improving Speech Emotion Recognition in Under-Resourced Languages via Speech-to-Speech Translation with Bootstrapping Data Selection**
- Hsi-Che Lin, Yi-Cheng Lin, **Huang-Cheng Chou**, Hung-yi Lee
- ICASSP 2025
- [DOI](#)
-
12. CO
- Emo-bias: A Large Scale Evaluation of Social Bias on Speech Emotion Recognition**
- Yi-Cheng Lin, Haibin Wu, **Huang-Cheng Chou**, Chi-Chun Lee, Hung-yi Lee
- INTERSPEECH 2024
- [DOI](#)

13. CO

EMO-Codec: An In-Depth Look at Emotion Preservation Capacity of Legacy and Neural Codec Models with Subjective and Objective Evaluations

Wenze Ren, Yi-Cheng Lin, Huang-Cheng Chou, Haibin Wu, Yi-Chiao Wu, Chi-Chun Lee, Hung-yi Lee, Hsin-Min Wang, Yu Tsao

APSIPA ASC 2024

[DOI](#)

Mentored

1. MENTORED

VoxEmo: A Reproducible Toolkit for Speech Emotion Recognition with Speech LLMs

Hezhao Zhang, Huang-Cheng Chou, Shrikanth Narayanan, Thomas Hain

IEEE Transactions on Affective Computing (under review); arXiv:2603.08936

[arXiv](#)

2. MENTORED

AdaLTM: Adaptive Layer-wise Task Vector Merging for Categorical Speech Emotion Recognition with ASR Knowledge Integration

Chia-Yu Lee, Huang-Cheng Chou, Tzu-Quan Lin, Yuanchao Li, Ya-Tse Wu, Shrikanth Narayanan, Chi-Chun Lee

arXiv:2603.25041, 2026

[arXiv](#)

3. MENTORED

Speaker-Aware Temporal Aggregation Strategies on Segment Representations for Depression Detection in Dyadic Interaction: A Benchmark Study

Anisha Pattanayak, Huang-Cheng Chou, Shrikanth Narayanan, Sudarsana Reddy Kadiri

arXiv:2607.02904, 2026

[arXiv](#)

4. MENTORED

Layer-wise Cross-Lingual Depression Detection from Speech: Analysis with Contrastive Alignment

Anisha Pattanayak, Hanie Kang, Huang-Cheng Chou, Shrikanth Narayanan, Sudarsana Reddy Kadiri

arXiv:2607.02920, 2026

[arXiv](#)

5. MENTORED

Can Conversational Temporal Dynamics Improve Depression Detection in Dyads? A Preliminary Investigation in Multi-Modality Perspectives

Hanie Kang, Huang-Cheng Chou, Sudarsana Reddy Kadiri, Shrikanth Narayanan

arXiv:2607.03744, 2026

[arXiv](#)

6.

MENTORED

Speaker Identity in Non-Verbal Vocalizations: Conditional Distillation and Mixture of Experts Approach

Tzu-Chieh Wei, Yi-Cheng Lin, **Huang-Cheng Chou**, Kuan-Yu Chen, Hsin-Yen Sung, Shrikanth Narayanan, Hung-yi Lee

INTERSPEECH 2026; arXiv:2606.21215

[arXiv](#)

7.

MENTORED

AMECxSV: Adaptive Metadata-Driven Embedding-Fusion Calibration for X-Lingual Speaker Verification

Xin Wei, Shi He, Yihe Yuan, **Huang-Cheng Chou**, Sudarsana Reddy Kadiri, Shrikanth Narayanan

IEEE SLT 2026 (submitted); arXiv:2607.16532

[arXiv](#)

Contact

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